Euclid Book I: Reconstruction
Proposition I.3
To Cut Off from the Greater of Two Given Segments a Segment Equal to the Less

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August 7, 2026

1 Introduction

Euclid's third proposition asks for the construction of a segment equal to a given shorter segment on a given greater segment.

Classically, this construction is obtained by combining Proposition I.2 with an additional circle construction. A segment equal to the shorter one is first laid off from an endpoint of the greater segment, and a circle is then used to determine the required point on the greater segment.

In the present reconstruction over Hilbert geometry, the logical structure is substantially different. The theory developed in Book Zero already contains an abstract notion of one segment being less than another. This notion is defined by the existence of a point between the endpoints of the greater segment such that the corresponding subsegment is congruent to the smaller one.

Consequently, the existential content of Euclid's Proposition I.3 is already embodied in the definition of segment inequality. The reconstruction therefore consists not in repeating Euclid's geometric construction, but in exposing the witness contained in the definition of `HilbertSegmentLess`.

This proposition provides the first clear example where a classical Euclidean construction becomes an immediate consequence of the abstract language developed in Book Zero.

2 The Classical Construction

Euclid's third proposition asks for the construction of a segment equal to a given shorter segment on a given greater segment.

Unlike Proposition I.2, the present construction does not introduce a new geometric technique. Instead, it combines Proposition I.2 with one additional circle construction.

The construction proceeds as follows.

2.1 Final Configuration

The objective is to determine a point E on the greater segment AB such that

$$AE \cong CD.$$

The final geometric configuration is shown in Figure 1.

From the construction we obtain

$$AE \cong CD.$$

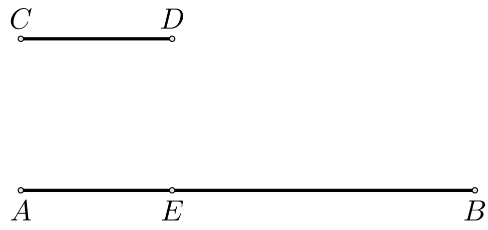

Figure 1: Final configuration of Euclid's construction for Proposition I.3.

The remaining task is to justify that the point E lies between the endpoints of the greater segment.

2.2 Construction Algorithm

The construction uses Proposition I.2 as an already established construction primitive.

2.2.1 Step 1. Given Segments

Let AB be the greater segment and CD the given shorter segment.

2.2.2 Step 2. Apply Proposition I.2

By Proposition I.2, construct a segment AF congruent to the given segment CD.

Thus

$$AF \cong CD.$$

2.2.3 Step 3. Draw the Circle Centered at A

Draw the circle centered at A passing through F.

2.2.4 Step 4. Determine the Required Point

Let E be the intersection of the circle with the greater segment.

Since

$$AE \cong AF$$

and

$$AF \cong CD,$$

it follows that

$$AE \cong CD.$$

The required segment has therefore been obtained.

3 Proof

The classical Euclidean proof constructs the required point by applying Proposition I.2 and then introducing an auxiliary circle.

The reconstruction over Hilbert geometry follows a different logical route. The relation that one segment is less than another is already represented in Book Zero by the predicate

$$\texttt{HilbertSegmentLess}\ CD\ AB.$$

By definition, this predicate means that there exists a point E between the endpoints of the greater segment satisfying

$$A - E - B$$

and

$$AE \cong CD.$$

Therefore, assuming

$$CD < AB,$$

we immediately obtain a point E such that

$$A - E - B \qquad \text{and} \qquad AE \cong CD.$$

This is precisely the conclusion required by Proposition I.3.

Consequently, over the present Hilbert/Book Zero layer, the proposition is an immediate consequence of the definition of `HilbertSegmentLess`.

□

4 Proof Architecture

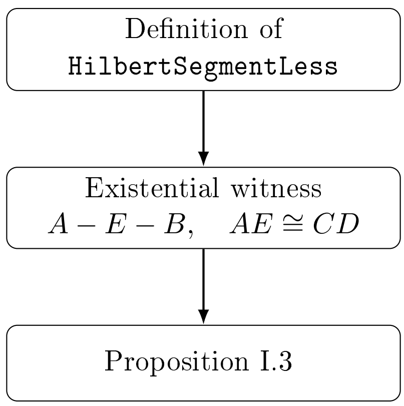

Figure 2: Logical structure of Proposition I.3 over the Book Zero layer.

5 Hilbert Reconstruction

Unlike Euclid's original proof, the present reconstruction does not repeat the geometric construction based on Proposition I.2.

The abstract notion of one segment being less than another has already been introduced in Book Zero by the predicate `HilbertSegmentLess`. By definition, this predicate contains exactly the existential information required by Proposition I.3.

Consequently, the reconstruction consists simply in extracting the witness from this definition.

This illustrates an important feature of the Book Zero layer. Classical geometric constructions are progressively replaced by abstract synthetic notions whose mathematical content has already been established.

6 Lean Formalization

The Lean formalization closely follows the mathematical argument.

The hypothesis

$$\texttt{HilbertSegmentLess}\ CD\ AB$$

already contains an existential witness providing the required point between the endpoints of the greater segment together with the corresponding congruence.

The formal proof therefore consists only of unpacking this witness and applying the symmetry of segment congruence.

Unlike the classical Euclidean proof, no additional geometric construction is required.

The resulting Lean theorem is therefore considerably shorter than the corresponding classical argument.